1. a) Write down the formal definition of $\Theta(g(n))$. [10%]

b) Simplify the following functions by expressing them in Θ-notation. Give a formal justification for each case, referring to the definition of Θ-notation.

(i) $n^3 + 2n^2 + 3n$

(ii) $2n - 5$

[30%]

c) Copy the following table to your answer booklet and indicate for each pair of functions $f(n), g(n)$ whether $f(n)$ is O, o, Ω, ω, or Θ of $g(n)$ by marking **all** cells that apply.

$f(n)$	$g(n)$	O	o	Ω	ω	Θ
n^3	n^3					
n^2	n					
$\log n$	n					
1000	n					
$\sum_{i=1}^{n} i$	n^2					
$\log_2 n$	$\log_{10} n$					

[30%]

d) The following algorithm turns all integers in an array $A[1 \ldots n]$ of length n into their absolute values.

```
ABSOLUTE-VALUES(A)
for i = 1 to A.length do
    if A[i] < 0 then
        A[i] = -A[i]
```

Prove the correctness of ABSOLUTE-VALUES by stating an appropriate loop invariant and showing the three properties: initialisation, maintenance, and termination.

[30%]

2. a) Your answers to the following questions should consider the following sorting algorithms: INSERTIONSORT, SELECTIONSORT, MERGESORT, HEAPSORT, QUICKSORT, RANDOMISED-QUICKSORT and BUBBLESORT.

(i) Name all algorithms that sort in time $O(n \log n)$ or expected time $O(n \log n)$. [10%]

(ii) Name all algorithms that have a worst-case time of $\Theta(n^2)$. [10%]

(iii) Name a sorting algorithm with a best-case time of $O(n)$. [5%]

b) Consider the execution of COUNTINGSORT on the following input array A. Recall that during the final for loop the input elements are written to an output array B based on counters stored in an array C whose contents just before the final loop are given below.

A	0	2	2	1	0	3	1

C	2	4	6	7

Copy the table below to your answer booklet and write down the contents of the arrays B and C at the end of each iteration of the final for loop.

Array C				Array B						
0	**1**	**2**	**3**	**1**	**2**	**3**	**4**	**5**	**6**	**7**

[20%]

c) (i) Explain how RADIXSORT can use COUNTINGSORT to sort n numbers with d digits, each digit taking values in $\{0, \ldots, k\}$. Give an argument as to why this approach is correct. [15%]

(ii) State the running time of RADIXSORT in the above setting. [10%]

d) Assume we want to find out whether, during a run of RANDOMISED-QUICKSORT on the following array, the two elements 2 and 4 are compared.

2	6	3	4	5	1

Explain what you can conclude about the decision whether to compare 2 and 4 considering the **first** call of RANDOMISED-PARTITION with the following choices of the pivot.

(i) The pivot is chosen as 4. [10%]

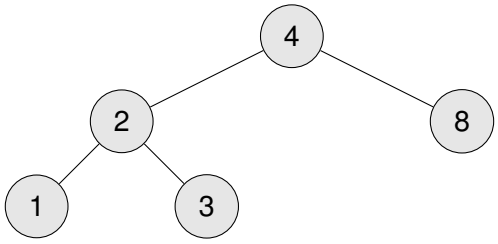
(ii) The pivot is chosen as 3. [10%]

(iii) The pivot is chosen as 5. [10%]

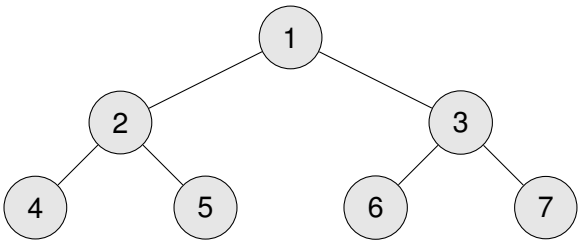
3. a) Give a recursive definition of binary trees. [10%]

b) A binary tree is *complete* if all leaves are on the same level. Prove by induction that every complete binary tree of height h has $2^h - 1$ internal nodes. [30%]

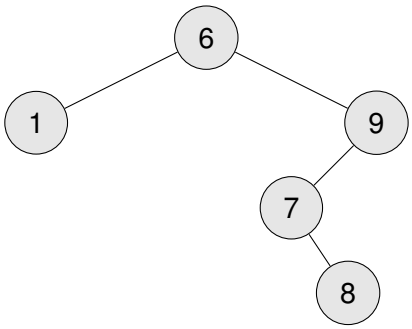
c) For each of the following trees, explain which show binary search trees.



(i) [10%]



(ii) [10%]



(iii) [10%]

d) Recall that the *height* of a node in a tree is the length of the longest path from that node down to a leaf. Assume a non-empty *full* tree, that is, all nodes are either leaves or have two children.

(i) State an equation that computes the height $h(v)$ for an internal node v from the height $h(v.\text{left})$ of its left subtree and the height $h(v.\text{right})$ of its right subtree. [10%]

(ii) Give an algorithm in pseudocode (or Java syntax) that recursively computes $h(v)$ for any node v in a non-empty full binary tree. [10%]

(iii) State the running time of your algorithm in Θ-notation when applied to the root of a full tree with n nodes. Justify your answer (briefly). [10%]

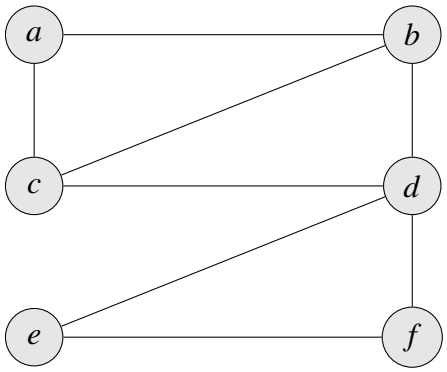
4. a) The following matrix shows the costs of constructing roads between five towns.

$$\begin{pmatrix} \infty & 1 & \infty & \infty & 6 \\ 1 & \infty & 4 & 2 & 7 \\ \infty & 4 & \infty & 3 & 8 \\ \infty & 2 & 3 & \infty & 9 \\ 6 & 7 & 8 & 9 & \infty \end{pmatrix}$$

(i) Draw a diagram of this graph (omit edges of ∞ cost). [10%]

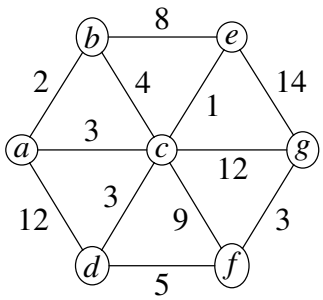
(ii) Find the cheapest road network connecting all towns and give its construction cost. State which algorithm you used. [20%]

b) Apply Breadth-first search starting at node a in the graph below. Assume that all adjacency lists are sorted alphabetically. Show the resulting tree and print the contents of the queue at each step that a node becomes black.



[30%]

c) Use Dijkstra's algorithm to work out the shortest path from a to g in the graph below. Show the working of the algorithm.



[30%]

d) Explain (briefly) why Dijkstra's algorithm is a greedy algorithm. [10%]

END OF QUESTION PAPER